

A critical radial-coarea gateway for generalized inverses

Partial progress on the endpoint conjecture in arXiv:2412.18976

Candidate solution packet

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Abstract

The source proves regularity of a generalized inverse for $f \in W^{1,p}$, $p > n - 1$, and conjectures the endpoint $p = n - 1$. The source's Morrey diameter estimate cannot survive critically. We prove instead that three explicit structural properties—multiplicity one on the positive-Jacobian set, projection-linking of inverse images of target segments, and a critical radial coarea inequality—imply the full desired $W^{1,1}$ and finite-distortion conclusion. The proof gives an L^1 -Poincaré inequality and isolates critical surface Lusin (N) as the remaining analytic obstruction. We also verify the full endpoint conclusion for all radial finite-distortion cavitation and collapsed-core maps.

1 Exact source problem

The source proves Theorem 1.2 for $p > n - 1$ and then states verbatim:

“We conjecture that it suffices to take $p = n - 1$ in Theorem 1.2. However, under this weaker assumption, one cannot conclude that h is differentiable almost everywhere.”

In higher dimensions, the inverse of a homeomorphism $f: \Omega \rightarrow \Omega'$ of finite distortion, belonging to $W_{\text{loc}}^{1,n-1}(\Omega, \mathbb{R}^n)$, also has finite distortion. In particular, $f^{-1} \in W_{\text{loc}}^{1,1}(\Omega', \mathbb{R}^n)$ (see [12]). We establish the following analogous result.

Theorem 1.2. *Let Ω and Ω' be domains in $\mathbb{R}^n$, $n \geq 3$ and $p > n - 1$, and let $f_0: \Omega \rightarrow \Omega'$ be a given homeomorphism. Suppose that a mapping $f \in W_{\text{loc}}^{1,p}(\Omega, \mathbb{R}^n)$ has finite distortion, satisfies the (INV) condition, and coincides with f_0 on a neighborhood of $\partial\Omega$. Then there exists a generalized inverse $h: \Omega' \rightarrow \mathbb{R}^n$ with finite distortion. In particular, $h \in W_{\text{loc}}^{1,1}(\Omega', \mathbb{R}^n)$.*

Moreover, the generalized inverse mapping $h: \Omega' \rightarrow \mathbb{R}^n$ is differentiable a.e. in Ω' and coincides with the classical inverse mapping a.e. on the set $f(\{x \in \Omega: J_f(x) > 0\})$. Additionally, the mappings satisfy $x \in h_T(f(x))$ for a.e. $x \in \Omega$ and $y \in f_T(h(y))$ for a.e. $y \in \Omega'$.

We conjecture that it suffices to take $p = n - 1$ in Theorem 1.2. However, under this weaker assumption, one cannot conclude that h is differentiable almost everywhere.

Here finite distortion is used in the source's sense: $J_f \geq 0$ and $J_f(x) = 0$ implies $Df(x) = 0$ a.e. Consequently $\text{adj } Df = 0$ a.e. on $\{J_f = 0\}$.

2 The three structural gateway properties

Put $m = n - 1$. Let G be a measurable subset of Ω on which $J_f > 0$, f is approximately differentiable, and f is one-to-one. Let $h \in L_{\text{loc}}^\infty(\Omega'; \mathbb{R}^n)$ be a proposed generalized inverse with $h(f(x)) = x$ for a.e. $x \in G$.

We isolate the following properties.

(M) Up to null sets, G contains $\{J_f > 0\}$ and $f|_G$ has multiplicity one.

(L) For a.e. $y_0, y_1 \in \Omega'$ and each coordinate projection $\pi_j : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$|h_j(y_1) - h_j(y_0)| \leq \mathcal{H}^1(\pi_j(f^{-1}([y_0, y_1]))) .$$

The inverse image is understood topologically at cavity values. This is the projection-linking conclusion obtained when nested topological images prevent a hyperplane from separating a fiber over a connected target set.

(C) For a.e. y_0 , every measurable $E \Subset \Omega$, and $q_{y_0}(x) = (f(x) - y_0)/|f(x) - y_0|$, one has

$$\int_{\mathbb{S}^{n-1}} \mathcal{H}^1(\pi_j(E \cap q_{y_0}^{-1}(z))) d\mathcal{H}^{n-1}(z) \leq \int_E |\operatorname{adj} Dq_{y_0}(x)| dx.$$

Property (C) is the radial coarea inequality. For a general critical Sobolev map, its null-set clause is not automatic. Csörnyei–Hencle–Malý prove it for endpoint homeomorphisms by first establishing Lusin (N) on almost every source hyperplane.

3 The gateway theorem

Theorem 1 (Critical radial-coarea gateway). *Let $n \geq 3$ and let $f \in W_{\operatorname{loc}}^{1,n-1}(\Omega; \mathbb{R}^n)$ have finite distortion. Suppose that a bounded generalized inverse h satisfies (M), (L), and (C) locally. Then $h \in W_{\operatorname{loc}}^{1,1}(\Omega'; \mathbb{R}^n)$. More precisely, there is $g \in L_{\operatorname{loc}}^1(\Omega')$ such that every ball $B = B(a, r) \Subset \Omega'$ satisfies*

$$\int_B |h - h_B| dy \leq C_n r \int_B g(y) dy. \quad (1)$$

The weak derivative of h vanishes a.e. outside $f(G)$ and satisfies $Dh(y) = [Df(h(y))]^{-1}$ a.e. on $f(G)$. In particular, h has finite distortion.

Proof. Fix $B = B(a, r) \Subset \Omega'$ and $y_0 \in B$ satisfying the gateway properties. If $y_1 \in B$ and $z = (y_1 - y_0)/|y_1 - y_0|$, then the segment $[y_0, y_1]$ lies in B . Property (L), followed by (C) on $E = f^{-1}(B)$, gives after integration on a target sphere centered at y_0

$$\begin{aligned} & \int_{B \cap \partial B(y_0, s)} |h(y) - h(y_0)| d\mathcal{H}^{n-1}(y) \\ & \leq C_n s^{n-1} \int_{f^{-1}(B)} |\operatorname{adj} Dq_{y_0}(x)| dx \\ & \leq C_n s^{n-1} \int_{f^{-1}(B)} \frac{|\operatorname{adj} Df(x)|}{|f(x) - y_0|^{n-1}} dx. \end{aligned}$$

The last inequality is the adjugate chain-rule estimate for radial projection. Integrating in s over the part of each ray contained in B and using $s \leq 2r$ yields

$$|h(y_0) - h_B| \leq C_n \int_{f^{-1}(B)} \frac{|\operatorname{adj} Df(x)|}{|f(x) - y_0|^{n-1}} dx.$$

Integrate this in $y_0 \in B$ and use

$$\int_B |z - y_0|^{1-n} dy_0 \leq C_n r \quad (z \in \mathbb{R}^n).$$

We obtain the critical Poincaré estimate

$$\int_B |h - h_B| dy \leq C_n r \int_{f^{-1}(B)} |\operatorname{adj} Df(x)| dx. \quad (2)$$

Finite distortion makes the integrand zero on $\{J_f = 0\}$. By (M), the area formula on the countable Lipschitz decomposition of the approximately differentiable set gives

$$\int_{f^{-1}(B)} |\operatorname{adj} Df|, dx = \int_B g(y) dy, \quad g(y) = \frac{|\operatorname{adj} Df(h(y))|}{J_f(h(y))} \mathbf{1}_{f(G)}(y).$$

Moreover, $g \in L^1_{\text{loc}}$ because its integral is bounded by $C_n \int |Df|^{n-1}$. Thus (2) is (1). The standard 1-Poincaré characterization of $W^{1,1}$ now gives $h \in W^{1,1}_{\text{loc}}$, with $|Dh| \leq C_n g$ a.e. In particular $Dh = 0$ off $f(G)$.

On each Lipschitz piece of G , the inverse differentiation identity gives $Dh(y) = [Df(h(y))]^{-1}$ for a.e. $y \in f(G)$. Hence $J_h > 0$ wherever $Dh \neq 0$, while both vanish off the good image. This is finite distortion. $\square$

Remark 1 (What remains for the original conjecture). *In the source's supercritical argument, strong (INV+) and nested topological images yield (L), while surface Lusin (N) yields (C) and is also used to obtain (M). At $p = n - 1$, continuous surface Sobolev maps can fail Lusin (N). Theorem 1 therefore removes the false critical Morrey step and shows that the unresolved issue is structural: prove these three consequences from finite distortion plus (INV), or construct a map where a critical exceptional surface creates singular inverse variation.*

4 An exact endpoint family

Proposition 1 (Radial cavitation and core collapse). *Let $f : B_R \rightarrow B_S$ be radial,*

$$f(x) = \rho(|x|) \frac{x}{|x|},$$

where ρ is nondecreasing and absolutely continuous away from the origin, $f \in W^{1,n-1}$, and f has finite distortion. Allow either a point cavity $\rho(0+) = b > 0$ or a collapsed core $\rho^{-1}(0) = [0, r_0]$. Then the natural radial generalized inverse is in $W^{1,1}(B_S; \mathbb{R}^n)$ and has finite distortion.

Proof. Where $\rho > 0$, the eigenvalues of Df are ρ' in the radial direction and ρ/r in the $n - 1$ tangential directions. Finite distortion forces $\rho' > 0$ a.e. there. Hence the inverse radius σ is absolutely continuous on (b, S) and the inverse derivative has eigenvalues $\sigma'(s)$ and $\sigma(s)/s$. Set $\sigma = 0$ in the cavity; at the single collapsed-core image point choose any value. Since $0 \leq \sigma \leq R$,

$$\begin{aligned} \int_{B_S} |Dh| &\leq C_n \int_b^S \left(\sigma'(s) + \frac{\sigma(s)}{s} \right) s^{n-1} ds \\ &\leq C_n \left(RS^{n-1} + \int_b^S \sigma(s) s^{n-2} ds \right) < \infty. \end{aligned}$$

The inverse eigenvalues are all positive off the constant cavity region and the exceptional target point. Thus the generalized inverse has finite distortion. $\square$

5 Limitations and novelty status

The theorem is conditional and does not prove that endpoint (INV) implies strong $(INV+)$, surface Lusin (N) , or a canonical fiber selection. A direct logarithmic-spike test shows that no local $W^{1,n-1}$ oscillation estimate can repair the source proof. Attempts to turn the source's slab jump example into a finite-distortion counterexample via a Ponomarev homeomorphism fail for topological reasons: a homeomorphism cannot map a null set onto the open bad slab. A subtler cell-like construction is not ruled out.

No later primary source located in bounded searches through August 13, 2026 claims the full endpoint. Priority is not asserted, and specialist review of the gateway hypotheses and the 1-Poincaré characterization is recommended.

References

- [1] A. Doležalová, S. Hencl, and J. Onninen, *(INV) condition and regularity of the inverse*, arXiv:2412.18976.
- [2] M. Csörnyei, S. Hencl, and J. Malý, *Homeomorphisms in the Sobolev space $W^{1,n-1}$* , J. Reine Angew. Math. 644 (2010), 221–235.
- [3] S. Hencl and P. Koskela, *Lectures on Mappings of Finite Distortion*, Lecture Notes in Mathematics 2096, Springer, 2014.